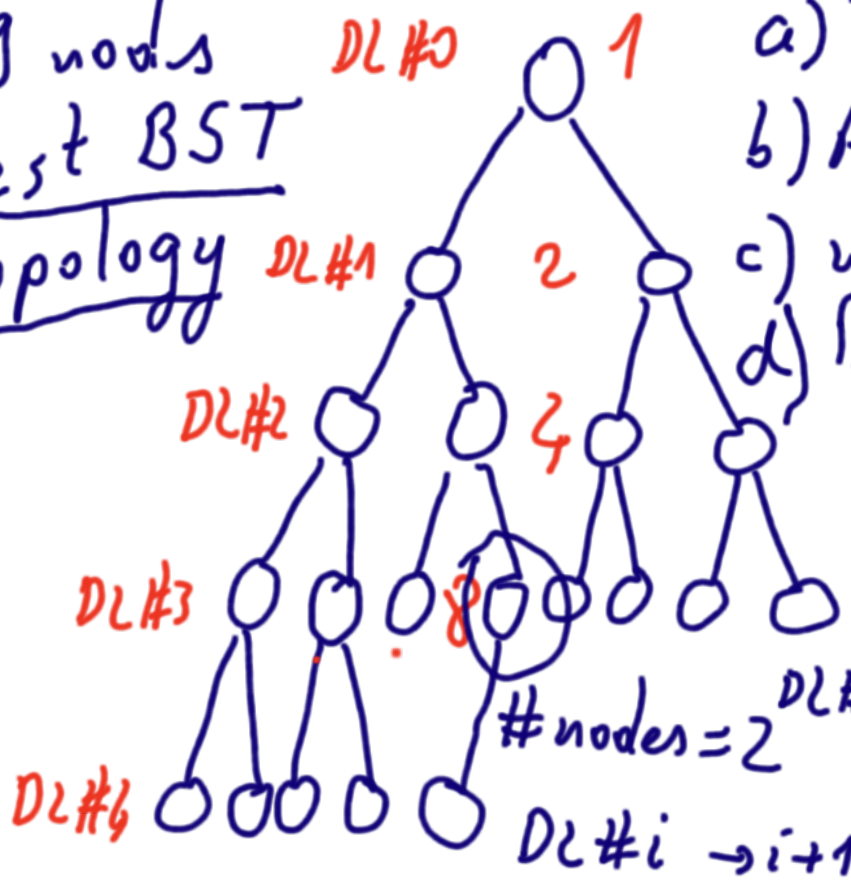
19 nods

Best BST

topology



a) Total access time

b) Average " "

c) Worst " "

d) Best " "

a) 69 "

b) 69/19

c) 5

d) 1

1

Q) n nodes in a BST with a best distribution of nodes Provide the distribution of nodes at depth levels?

DL#	0	1	2	3	4	...	i	...	k
#nodes	1	2	4	8	16	...	2^i	...	2^k ?

x: # nodes at the last DL

$x = n - (2^{k+1} - 1)$; $k = ?\ f(n)$ in terms of n only

e.g.) for $\forall n \in \{16, \dots, 31\}$ $2^{k+1} - 1 = 15$

$\Rightarrow 2^{\lfloor \log_2 n \rfloor} - 1 \Rightarrow k+1 = \lfloor \log_2 n \rfloor$

$\Rightarrow x = n - [2^{\lfloor \log_2 n \rfloor} - 1]$

6